

The disadvantage of this formulation is that the risk term in the objective function is expressed in terms of variance and may require several iterations to determine the trade schedule corresponding to the investor's level of risk aversion. For large trade lists this problem can be quite resource taxing. Malamut (2002) provided an adjustment to the QP model to directly convert the standard deviation risk aversion parameter to the variance risk aversion parameter which can be used to further simplify the risk aversion/variance aversion issue.

Trade Schedule Exponential

The trade schedule exponential approach parameterizes the trade schedule based on an exponential decay function with parameter θ_i . It is a non-linear optimization routine that uses the square root function for our risk expression.

The number of stocks to transact in a period is determined as follows:

$$x_{ij} = X_i \cdot \frac{e^{-j\theta_i}}{\sum_{k=1}^n e^{-k\theta_i}} \quad (9.8)$$

The optimization formulation for the trade schedule exponential approach is a non-linear optimization formulation:

$$\underset{x}{Min} \sum_{i=1}^m \sum_{t=1}^n \frac{b_1 \cdot I_i^* \cdot x_{it}^2}{X_i \cdot v_{it}} + \lambda \cdot \sqrt{\sum_{k=1}^n r_k' C r_k} \quad (9.9)$$

Subject to:

$$\begin{aligned} x_{ij} &= X_i \cdot \frac{e^{-j\theta_i}}{\sum_{k=1}^n e^{-k\theta_i}} && \text{for all } i, j \\ r_{ij} &= \sum_{k=j}^n x_{ik} && \text{for all } i, j \\ LB_i &\leq \theta_i \leq UB_i && \text{for all } i \end{aligned}$$

Expressing the trade schedule as a parametric exponential formulation provides many benefits. First, there is only one parameter to estimate for each stock regardless of the number of specified trading periods and trading days. For example, an m-stock portfolio executed over n trading horizons has only m parameters to determine regardless of the trading horizon whereas the complete problem and QP optimization has n*m decision variables. Second, since Equation 9.9 is true for all stocks we are guaranteed of completion. Third, since $e^{-j\theta_i} > 0$ for all j we have $x_{ij} > 0$ for all periods and are ensured to adhere to the shrinking portfolio constraint.